A Kind of Effective Method of Removing Compound Noise in Image

Hongyan Chen
Electronic Engineering College,
Chengdu University of Information Technology
Chengdu, China

Abstract—**The noise is generally not a single type in digital image, which is usually composed of Gaussian noise and pulse noise. So the signal to noise ratio of the image is low and the noise in images needs to be processed. The traditional denoising algorithm and various improved algorithm is effective for single Gaussian noise or impulse noise mostly, these denoising algorithms are unable to obtain satisfactory denoising effect in dealing with the mixed noise. Some existing hybrid denoising methods usually change distribution characteristics of another noise when removing one noise at the same time, so subsequent denoising effect is affected. To resolve the problem, a composite image denoising method based on the combination of independent component analysis and traditional spatial denoising method is proposed. Simulation experiment results show that the method can effectively preserve edge details and other information of image when removing mixed noise.**

Keywords-gaussian noise; pulse noise; compound noise; image processing; independent component analysis

I. INTRODUCTION

In recent years, with the popularity of digital devices, digital image is becoming more and more important for people's daily life, production and so on. But because of the influence of mechanical and electrical properties of digital equipment and the imaging environment, the image accessed directly is often polluted by many kinds of noise, such as composite type noise composed of Gaussian noise and impulse noise[1]. So subsequent image processing effect such as image segmentation, edge detection is affected by the noise. In some remote control situation, even to the extent that some important information is not fed back to the user accurately. These probably result in an error determination and cause unnecessary losses.

At present, there are many methods of filtering Gaussian noise and impulse noise. Mean filter, median filter, Wiener filter, wavelet transform and some other filtering methods are commonly used[2][3][4][5][6][7][8]. While these traditional algorithm and various improved algorithm on the basis of them are for single noise. For Gaussian noise and impulse noise in images, these methods are unable to be obtained when using these algorithms to process mixed noise of Gaussian and impulse. In recent years, some scholars put forward the denoising method for compound noise in digital images[9] [10]. One way of thinking is to use the separate pixel method to distinguish the pixels polluted by different noise, and then use the median filter or the neighborhood average method to remove noise. Another idea is to use one effective method to remove one of the noise to get the transitional image first, and then use another effective way to deal with the other noise in transitional image. However, these hybrid denoising method will change the distribution characteristics of the other noise in transitional image while removing one kind of noise, which will affect the subsequent denoising effect. For example, there are Gaussian noise and pulse noise in an image, if we get transitional image by removing pulse noise first, then the noise in transitional image is no longer Gaussian noise usually. So we can't get satisfactory effect finally when the transitional image are processed by algorithms used to remove Gaussian noise. In view of the problems and the advantages of various filtering methods, a combined denoising method based on independent component analysis[11][12] and traditional spatial denoising method is proposed in the paper.

II. ALGORITHM PRINCIPLE

A. Gaussian noise filtering method based on independent component analysis

The noise model of independent component analysis is shown as follows：

$$X = A(S+N) = \sum_{i=1}^{m} a_i(s_i + n_i), i = 1,2,...m \qquad (1)$$

$X = [x_1, x_2, ..., x_m]^T$ is mixed signal by unknown factors, that is observed signal. $S = [s_1, s_2, ..., s_n]^T$ is the original signal($m \geq n$). Each component of S is statistically independent. N is Gaussian noise which is independent from S (Assumed that noise covariance matrix is $\sigma^2 I$), A is mixing matrix.

This work was supported in part by Education Department Foundation of Sichuan Province of China under Grants 15ZB0171 and Youth Scholar Foundation of Chengdu University of Information Technology of China under Grant CRF201402

Authorized licensed use limited to: University of Zagreb Faculty of Electrical Engineering and Computing. Downloaded on March 27,2026 at 13:53:04 UTC from IEEE Xplore. Restrictions apply.

ICA transform is essentially to search for the appropriate solution of the mixed matrix W based on the above formula and the component s_i in Y ($Y = WX$) is as statistically independent as possible. So the noise variance in the mixed components $Y = WX$ is still $\sigma^2 I$. That is, the component y_i is signal s_i (to be estimated) and Gaussian random variables n_i (the variance is σ^2). For the image data, s_i in the transformed component y_i has strong non Gaussian property. That is, the majority of the elements in the transformed components s_i is zero, with has sparse distribution characteristics. Therefore, each component y_i is used to carry out the contraction operation and the estimated value of s_i is obtained by s_i^*[13][14][15][16]:

$$s_i^* = f(y_i) = sign(y_i)\max(0, |y_i| - \sqrt{2}\sigma^2) \qquad (2)$$

So the noise in y_i will be reduced by above operation. Then the denoising data is obtained[6] when the data s_i^* is projected back to the original image space.

The procedure of algorithm based on independent component analysis is as follows:

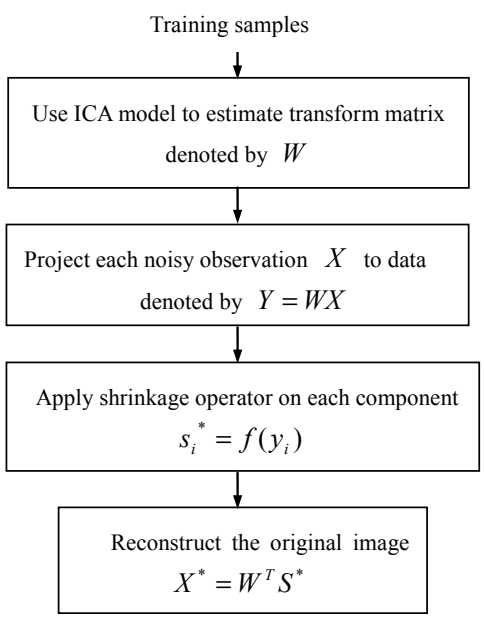

Figure 1. The flow block of Gaussian noise filtering method based on independent component analysis

Use a noise free training set to estimate mixing matrix A and orthogonal the obtained matrix A firstly. Then denote the orthogonal mixing matrix by W^T . Compute for each observed data X the corresponding noisy sparse components by the formula $Y = WX$. Apply the shrinkage nonlinearity function $f(\cdot)$ as defined above on each component y_i and denote the obtained components by s_i^*.

Finally, invert the data S^* to the original image space to obtain estimates of the noise free data by $X^* = W^T S^*$ .the noise free data is needed to estimate weight matrix W in advance when the proposed algorithm used in image denoising. It is not realistic in practical application. So we choose some noise free image with the same statistics to estimate weight matrix if the noise image is constrained to be natural.

B. Impulse noise filtering method based on spatial domain

Median filtering is to replace the current pixel value by the gray value of the adjacent pixels of it, that is,

$$f(x, y) = \underset{(s,t)\in s_{xy}}{median}\{g(s,t)\} \qquad (3)$$

The median filtering method is superior to other spatial filtering methods when it is used to process impulse noise in image. But the filtering effect is affected by the size of the filter window. In order to improve the denoising effect, the adaptive median filter is used to suppress the impulse noise. The method can reduce the distortion of edges in image.

Order $z_{\max}$, z_{med} , $z_{\max}$ is the gray level of the minimum, median and maximum value respectively, order z_{xy} is the gray value of the coordinates (x, y) , $s_{\max}$ is the maximum size allowed of s_{xy} .Adaptive median filter works at two levels:

- level Ⅰ : $A_1 = z_{med} - z_{\min}$, $A_2 = z_{med} - z_{\max}$, if $A_1 > 0$ and $A_2 < 0$, turn to Ⅱlelel, Otherwise increase the size of the window; if size of the window $\le s_{\max}$, duplicate level Ⅰ , otherwise output z_{xy} 。
- level Ⅱ : $B_1 = z_{xy} - z_{\min}$, $B_2 = z_{xy} - z_{\max}$, if $B_1 > 0$ and $B_2 < 0$, output z_{xy} , otherwise output z_{med} 。

C. Compound denoising method

For compound noise of Gaussian and pulse in image, the method of compound is used to remove these two kinds of noises. Firstly, transitional image is obtained by the method for one kind of noise, and then the image is processed by the method for another kind of noise. The processing flow is shown in figure 2.

If the distribution characteristics of another noise is changed after the elimination of the noise I, then the noise in the transitional image will no longer be noise II. Obviously, denoising effect will be affeced finally when use method for noise II to process transitional image. Therefore, the method

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 27,2026 at 13:53:04 UTC from IEEE Xplore. Restrictions apply.

based on the independent component analysis is used to remove the Gaussian noise. According to the analysis of Section A, the contraction operation of component y_i in ICA domain just remove Gaussian noise(n_i) in y_i and the distribution characteristics of other data is not changed. Then when the estimate valuse(s_i^*) of y_i is transformed back to the original image space, the effect of Gaussian noise is just eliminated and the distribution characteristics of pulse is not changed in image. Therefore, the image is processed firstly by the method based on ICA to eliminate the Gaussian noise, and then the adaptive median filter is used to eliminate the impulse noise in the paper.

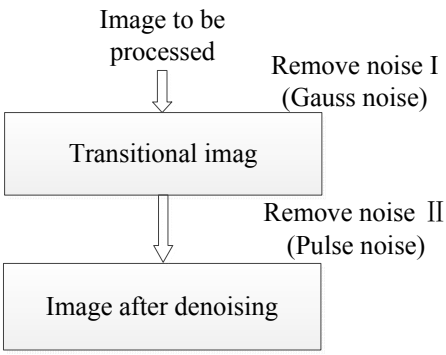

Figure 2. The flow block of the algorithm proposed in the paper

Because the image data is processed here, the sliding window is used in processing the two kinds of noise. For example, the specific algorithm steps are as follows when processing Gauss noise:

1) Obtain the weight matrix of sparse coding denote by W . First, we choose six natural images shown in Figure 3, from which a set of subwindows of 8×8 pixels are intercepted at random. Such windows were presented as 64-dimensional vectors of gray-scale values. There are 5000 samples of the vectors and the training data are in the form of 64×5000. Then the fast fixed-point algorithm based on symmetric orthogonality (FastICA)[17][18][19][20] is applied to learn the training sample and get the basic vectors after convergence of the FastICA algorithm. Finally, we get mixing matrix denoted by A whose size is 64×64 pixels. Each column of A is a vector of 1×64. Here we orthogonalize the mixing matrix A and denote the orthogonal matrix by W^T .

2) Project each noisy data X to other data space by formula $Y = WX$

3) compute for each y_i by the formula $s_i^* = f(y_i)$ to eliminates the Gauss noise whose value is smaller in higher-order information.

4) Project the noise free data S^* back to the original image space to recover original images by $X^* = W^T S^*$.

It is worth noting that the above processing was done separately for each overlapping subwindow in the image, 64 reconstructions were obtained for each pixel. The final reconstruction was obtained as the mean of the 64 reconstructions.

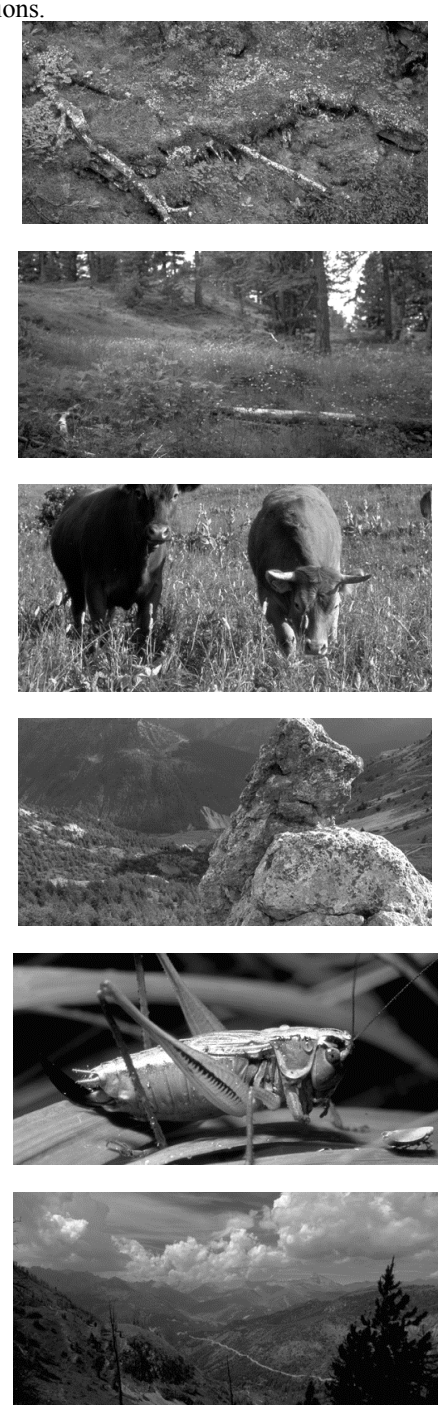

Figure 3. Six natural images selected to get training data

III. SIMULATION EXPERIMENT ANALYSIS

In order to analyze the effect of the compound denoising method proposed in paper, the image named "lena" which is

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 27,2026 at 13:53:04 UTC from IEEE Xplore. Restrictions apply.

shown in Figure 4 is selected from website http:/www.cis.hut.fi/projects/ica/data/images to be testing image.

Figure4.
Original image

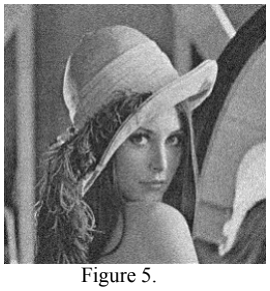

Figure 5.
Image with Gaussian noise

Figure6.
Image with compound noise

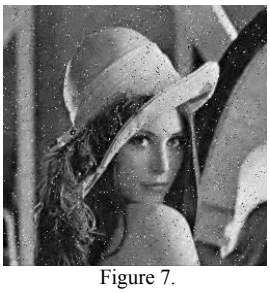

Figure 7.
Transitional image

Figure 8.
Image after denoising compound noise

Figure 9.
Median filtered image

Figure 10. Mean filtered image

The original image is polluted by Gaussian noise whose variance is 0.005 is shown in figure 5. Figure 6 is the image polluted by the pulse noise (density is 0.02) and the Gaussian noise (variance is 0.005). Figure 7 is the transitional image which is obtained by the method in this paper, that is, it is considered to be contaminated by impulse noise only. In fact, it can be clearly seen from Figure 7 that there are pulse noise in image. According to the analysis of the second Ⅱ, the shrinkage operation reduce Gaussian noise only and distribution characteristic of other data is not changed in the processing . So , we think that the noise in the transitional image is still pulse noise and we can select a method specifically aimed at pulse noise to process the transitional image. Then the transitional image is processed by adaptive median filtering and the denoising image is shown in Figure 8. In order to compare the denoising effect, the image in figure 6 is processed only by adaptive median filtering and the denoising image is shown in Figure 9. It can be seen that there are still some noise in the image and the useful information of the image is submerged in noise. This phenomenon is more obvious in smooth region although edge and details are retained well.The image in figure 6 is processed only by mean filtering and the denoising image is shown in Figure 10. we can see that the edge and details is so fuzzy although many noise is smoothed out. It can be seen that the edge and detail information is preserved effectively and the distribution characteristics of pulse noise is maintained while suppress the Gaussian noise in original image by comparison of Figure 4 and figure 7. The method used in this paper can suppress the Gaussian and impulse noise effectively by comparison of Figure 4 and Figure 8. It also can be seen that the quality of image processed by the method proposed is better than image processed by the single tranditional method. The effect of edge and texture details remained are also better than single tranditional method by comparison Figure 8, figure 9 and Figure 10. At the same time, we can see from figure 9 and figure 10 that the distribution characteristics of the rest noise in them is not determined accurately. The detailed information of the original image in them is also lost. So There is no sense in the subsequent processing of these two images.

IV. CONCLUSION

The noise distribution in the image acquired is complex and generally contains many kinds of noise in the practical applications. So the noise can't be effectively removed by a single denoising method. And it is difficult to separate the pixels which are contaminated by different noise by method of pixel separation. So another way of thinking is adopted. That is , an effective method is used to remove one of the noise to get the transition image first, and then the image is processed by an effective method to deal with another noise. In order to get better effective result, we are required to know the distribution characteristics of the other noise in the transition image. It is obviously very difficult to meet this requirement. The compound method based on the independent component analysis and the adaptive median filtering is an effective denoising method and the proposed method meets the requirement. Firstly, the transitional image is obtained by shrinkage operation, then the transitional image is processed by adaptive median filtering. The processing method and the processing order make full use of the advantages of the method for the single noise. It also retains

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 27,2026 at 13:53:04 UTC from IEEE Xplore. Restrictions apply.

original distribution characteristics of noise in transitional image. So it is an effective method to filter compound noise.

REFERENCES

[1] CHen hua, CHen Ting, Xie Min, “Wavelet packet analysis based on the two threshold denoising image restoration method, ”Optical technology,2008,34(2):215-217.

[2] Tian Pei,Li Qing-zhou,Ma Ping,Niu Yu-guang, “A new method of image denoising based on Wavelet Transform, ”Chinese Journal of image and graphics,2008,13(3):394-399.

[3] Preety D. Swami, Alok Jain, “Image denoising by supervised adaptive fusion of decomposed images restored using wave atom, curvelet and wavelet transform , ”Signal, Image and Video Processing, vol.8, pp.443-459, 2012.

[4] Leipo Yan, L.P. Wang and Kim Hui Yap, “A noisy chaotic neural network approach to image denoising, ” IEEE International Conference on Image Processing (ICIP 2004), Singapore, October, 2004.

[5] G. Motta, et al, “The iDUDE framework for grayscale image denoising," IEEE Trans. Image Processing, Vol.20, pp.1-21, 2011.

[6] Wei Zhu and Tony Chan, “Image Denoising Using Mean Curvature of Image Surface, ”SIAM J. Imaging Sci., vol.5, pp.1-32, 2012.

[7] L. P. Wang, Leipo Yan, and Kim Hui Yap, “Image denoising using stochastic chaotic simulated annealing, ” in Intelligent Multimedia Processing with Soft Computing, Yap Peng Tan, Kim Hui Yap, and L. P. Wang (eds.), Springer, Berlin, pp. 415-429, 2005.

[8] Zhang Jia-cheng,Fan Yong, “Image impulse noise filtering algorithm. Computer engineering and design, ”2010, 31(17): 3845-3847.

[9] Yang Bo,Geng Bo-ying , “Gaussian pulse mixed noise removing method for image matching recognition, ” Sensors and micro systems,2012,31(6):8-11.

[10] Jiang CHun-miao,Zhou Zuo-feng, “An effective method to remove Gaussian impulse noise from images, ”Computer engineering and Applications,2009,45(24):183-185.

[11] Hyvarinen A, et al. Independent Component Analysis. New York : John Wiley, 2001

[12] CHen Zai-ping,Gu Xiu-ping, ZHang Jing-lei, “mage denoising method based on independent component analysis (ICA) , ” Journal of Tianjin University of Technology,2009,25: 57-60.

[13] CHen Hong-yan,Fu Yu-sheng,Zhang Rui,Pi Yi-ming, “Speckle reduction of polarimetric SAR images based on ICA-SCS algorithm, ” Journal of electronics and information, 2007, 4: 819-821.

[14] L Shang,CH Zheng, “Image Feature Extraction and Denoising Based on Sparse Coding, ”Acta Simulata Systematica Sinica,2005.

[15] Y Tang，Y Chen，N Xu，A Jiang，L Zhou, “Image denoising via sparse coding using eigenvectors of graph Laplacian, ”Digital Signal Processing, 2016, 50(C):114-122.

[16] H Zhao，L Zhang, “Sparse Coding Image Denoising Based on Saliency Map Weight, ”Neural Information Processing-international Conference,2011,7063:308-315.

[17] JB Tang，QN Zeng, “W Guo, A Fast Fixed-Point Algorithm Based on ICA, ” Audio Engineering, 2007.

[18] ZL Ruan,LP Li,GB Qian，MG Luo, “Fast fixed-point algorithm based on complex ICA signal model with noise, ”Dianzi Yu Xinxi Xuebao/journal of Electronics ＆Information Technology, 2014, 36(5):1094-1099.

[19] W Yuan,LY Zhang , “Blind Image Separation Based on an Optimized Fast Fixed Point Algorithm, ” Advanced Materials Research , 2013, 756-759(756-759):3578-3583.

[20] J Thomas,Y Deville,S Hosseini, , “Time-domain fast fixed-point algorithms for convolutive ICA, ” IEEE Signal Processing Letters, 2006, 13(4):228-231.

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 27,2026 at 13:53:04 UTC from IEEE Xplore. Restrictions apply.